

Highlights

Six-degree-of-freedom cascaded fractional-order control of an over-actuated ROV: margin-constrained tuning and a bias-controlled comparison of metaheuristics

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- Cascaded FOPID-(1+TFOID) control of all six DOF of an over-actuated BlueROV2 Heavy
- Phase margins are imposed over a range of thruster lag and latency during tuning
- T200 thrust and electrical power are fitted to the manufacturer's measurements
- Equal budgets and a mirrored search box expose the origin bias of SOO and GJO
- The fractional cascade is the most robust: no divergence in 100 Monte Carlo runs

Six-degree-of-freedom cascaded fractional-order control of an over-actuated ROV: margin-constrained tuning and a bias-controlled comparison of metaheuristics

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Abstract

Fractional-order controllers for remotely operated vehicles are mostly designed for one axis, with idealized thrusters and no latency. We recast the cascaded FOPID-(1+TFOID) controller, proposed for load-frequency control, as a position-velocity cascade for all six degrees of freedom of an over-actuated BlueROV2 Heavy with eight T200 thrusters, whose thrust, dead band, saturation and electrical power are fitted to the manufacturer's measurements. Because the thruster dynamics are uncertain, tuning minimizes a multi-scenario six-DOF ITAE and energy cost subject to phase margins on every linearized loop at a nominal and two extreme thrusters (lag 0.05–0.2 s, latency 10–60 ms). Six metaheuristics, including the recent stellar oscillation optimizer (SOO), are compared at an equal evaluation budget and in a mirrored search box. On shifted benchmarks, SOO, golden jackal and grey wolf optimization are attracted to the coordinate origin; on the tuning problem, particle swarm optimization gives the lowest median cost (40.0, unchanged in the mirrored box) and SOO a significantly higher one (85.0, $p = 0.002$). The

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proposed cascade does not beat a tuned integer P-PID cascade on the nominal training cost (best 35.8 against 33.4), but it is the most robust structure: lowest error in the multi-set-point storm test (128.8 against 152.8), 6.5 times lower error than P-PID under severe sensor noise, and no divergence in 100 Monte Carlo draws of plant and actuator parameters (P-PID: 8). Tuning without latency and margin constraints gives margins down to 3.8° and a storm-test cost 1.8–5.6 times higher at 30 ms of latency.

Keywords: remotely operated vehicle, fractional-order control, cascade control, FOPID, tilt-integral-derivative, metaheuristic tuning, phase margin, BlueROV2

1. Introduction

[TODO: Paragraph 1: ROV/AUV motion control, coupled nonlinear hydrodynamics, wave and current disturbances; most FO studies regulate a single axis.]

[TODO: Paragraph 2: our earlier study [1]: surge-axis $PI^\lambda D^\mu$ [2], fourth-order plant identified from T200 measurements, PSO [3] and DE [4] over a multi-scenario cost, four tests, FPGA realization. Limits: one axis, no coupling, no allocation, no latency or margin requirement.]

[TODO: Paragraph 3: cascaded FO controllers in load-frequency control; FOPID-(1+TFOID) of [5] with SOO [6]; its own stated limits (no formal stability analysis, simulation only).]

[TODO: Paragraph 4: tuning by new metaheuristics; equal-iteration comparisons that give some methods twice the evaluations; origin/centre bias [7, 8]; inconsistencies between the SOO paper and its code [9].]

The contributions of this paper are:

1. The FOPID-(1+TFOID) cascade is recast as a position–velocity cascade (the second stage acts on $u_1 - \nu$, the mechanical counterpart of $u_1 - \Delta f$) and applied to all six DOF of an over-actuated BlueROV2 Heavy, with six benchmark structures realized in the same cascade form.
2. Tuning minimizes a multi-scenario cost (six-DOF ITAE and electrical energy) subject to phase margins on every linearized DOF loop, including the command latency and the thruster lag. Because the T200 dynamics are not known well, the margins are imposed over a range of thruster lag and latency. Without the constraint, ITAE tuning returns loops with margins below 10° that one or two extra samples of latency destabilize (Sect. 6.5).
3. The thruster model (thrust, dead band, asymmetry, saturation and electrical power at 10–20 V) is fitted to the manufacturer’s measurements [10], so the effort term is energy in joules and battery voltage is a physically meaningful uncertainty.
4. Six tuners, including the newest (SOO), are compared at an equal evaluation budget, in the original and in a mirrored search box, and on shifted benchmark functions; this separates search ability from attraction to the coordinate origin.
5. The four tests of [1] are extended to six DOF and complemented by plant, actuator, latency and thruster-loss variations and a Monte Carlo study.

Table 1: BlueROV2 Heavy parameters [12].

mass m	13.5 kg	volume ∇	0.0135 m ³ (neutral)
I_x, I_y, I_z	0.26, 0.23, 0.37 kg m ²	$\mathbf{r}_b$	[0, 0, -0.01] m
added mass	$X_{\dot{u}}$ 6.357, $Y_{\dot{v}}$ 7.121, $Z_{\dot{w}}$ 18.69 kg; $K_{\dot{p}}$ 0.1858, $M_{\dot{q}}$ 0.1348, $N_{\dot{r}}$ 0.2215 kg m ²		
linear damping	X_u 13.7, Y_v 0, Z_w 33 N s/m; K_p 0, M_q 0.8, N_r 0 N m s/rad		
quadratic damping	$X_{u u }$ 141, $Y_{v v }$ 217, $Z_{w w }$ 190 N s ² /m ² ; $K_{p p }$ 1.192, $M_{q q }$ 0.47, $N_{r r }$ 1.5 N m s ² /rad ²		

2. Vehicle and actuator model

2.1. Rigid-body and hydrodynamic model

Let $\boldsymbol{\eta} = [x, y, z, \phi, \theta, \psi]^\top$ be the position and attitude in the north-east-down frame and $\boldsymbol{\nu} = [u, v, w, p, q, r]^\top$ the velocities in the body frame, whose origin is the centre of gravity. The vehicle obeys [11]

$$\begin{aligned} \mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{g}(\boldsymbol{\eta}) &= \mathbf{T}\mathbf{f} + \boldsymbol{\tau}_d, \\ \dot{\boldsymbol{\eta}} &= \mathbf{J}(\boldsymbol{\eta})\boldsymbol{\nu}, \end{aligned} \quad (1)$$

where $\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$ is diagonal, $\boldsymbol{\nu}_r = \boldsymbol{\nu} - [\mathbf{R}^\top \mathbf{v}_c; \mathbf{0}]$ is the velocity relative to an irrotational current $\mathbf{v}_c$ (the term $\mathbf{M}_A \dot{\mathbf{v}}_c$ is neglected), $\mathbf{D}(\boldsymbol{\nu}_r) = \mathbf{D}_l + \mathbf{D}_q \text{diag}(|\boldsymbol{\nu}_r|)$, and $\mathbf{g}$ contains the restoring forces of the weight and buoyancy. The parameters of the heavy configuration are those of the open benchmark of [12] (Table 1).

2.2. Thrusters and allocation

Four horizontal thrusters at 45° and four vertical thrusters give the 6×8 allocation matrix $\mathbf{T}$ of [12]. A demanded generalized force $\boldsymbol{\tau}$ is distributed by $\mathbf{f}_d = \mathbf{T}^+ \boldsymbol{\tau}$ and scaled down as a whole when a thruster limit is reached, so the direction of $\boldsymbol{\tau}$ is kept. Each command is mapped to a normalized PWM

value $u_i \in [-1, 1]$ by the inverse of the nominal 16 V curve, with dead-band compensation. The thrust produced is

$$F(u) = \begin{cases} \sum_{k=1}^3 a_k (u - d_+)^k, & u > d_+, \\ 0, & -d_- \leq u \leq d_+, \\ -\sum_{k=1}^3 b_k (-u - d_-)^k, & u < -d_-, \end{cases} \quad (2)$$

followed by a first-order lag τ_m . No reliable dynamic model of the T200 is available: the manufacturer estimates 25–40 ms of ESC latency for a running motor, and about 110 ms has been reported from standstill. We therefore take $\tau_m = 0.1$ s and a latency of 30 ms as nominal values and design for $\tau_m \in [0.05, 0.2]$ s and latencies of 10–60 ms (Sect. 4). The coefficients were fitted by least squares to the manufacturer’s measurements at 10–20 V [10]; at 16 V the fit error is 0.24 N RMS (1.3 N maximum) over the range -39.9 N to 51.4 N. The electrical power of the actual thrust, $P(F) = \sum_k q_k |F|^k$, is fitted to the same data (3.1 W RMS of 389 W). The command reaches the thrusters three control periods ($3\Delta t = 30$ ms) after the measurement it was computed from.

2.3. Disturbances and navigation errors

Waves act as a sum of five sinusoids per DOF with random frequencies in 0.4 rad/s to 1.6 rad/s (amplitudes 20, 20, 15 N and 1 N m), and a current of 0.25 m/s ($\pm 20\%$ slow variation) acts through $\boldsymbol{\nu}_r$. The controller sees the output of a navigation filter, modelled as first-order Gauss–Markov errors with 2 Hz bandwidth (one sigma: position 1 cm, depth 5 mm, attitude 0.17–0.29°, velocity 1 cm/s, rates 0.29°/s).

3. Controller

3.1. Position-velocity form of the FOPID-(1+TFOID)

For each DOF i , the first stage acts on the position error $e_{1,i}$ (expressed in the body frame for the translational DOFs) and returns a velocity reference,

$$u_{1,i}(s) = \left(K_p + \frac{K_{i1}}{s^{\lambda_1}} + K_{d1}s^{\mu_1} \right) e_{1,i}(s). \quad (3)$$

The second stage acts on $e_{2,i} = u_{1,i} - \nu_i$ and returns an acceleration demand,

$$a_i(s) = \left(1 + \frac{K_t}{s^{1/n}} + \frac{K_{i2}}{s^{\lambda_2}} + K_{d2}s^{\mu_2} \right) e_{2,i}(s), \quad \tau_i = M_{ii} a_i. \quad (4)$$

In [5] the second stage acts on $u_1 - \Delta f$; here the local feedback is the measured velocity, which makes the cascade a position-velocity cascade in the usual sense of marine control. Scaling by the nominal inertia M_{ii} makes the gains dimensionally consistent across DOFs. One parameter set is shared by (x, y, z) and one by (ϕ, θ, ψ) : 22 decision variables.

3.2. Benchmark structures

Table 2 lists the six benchmarks. FOPID is the single-loop $\text{PI}^\lambda \text{D}^\mu$ of [1] extended to six DOF, P-PID the integer cascade of common ROV autopilots, and the three fractional cascades are those compared in [5]. All cascades use the position-velocity form of Eqs. (3)–(4), so the comparison isolates the stage transfer functions.

3.3. Realization

Each s^α is approximated by the Oustaloup filter [13] of order $N = 5$ on 10^{-3} rad/s to 10^2 rad/s ; for $\alpha < -1$ an exact integrator is split off. Every first-order section is discretized by the Tustin rule at $\Delta t = 10 \text{ ms}$. Integer

Table 2: Controller structures. G_1 acts on the position error; G_2 on $u_1 - \nu$ (cascades only).

Decision variables: two groups (translation, rotation).

structure	$G_1(s)$	$G_2(s)$	variables
FOPID-(1+TFOD) (proposed)	$K_p + K_{i1}s^{-\lambda_1} + K_{d1}s^{\mu_1}$	$1 + K_t s^{-1/n} + K_{i2}s^{-\lambda_2} + K_{d2}s^{\mu_2}$	22
FOPID [1]	$K_p + K_{i1}s^{-\lambda} + K_{d1}s^{\mu}$	–	10
PID	$K_p + K_{i1}/s + K_{d1}s/(s/\omega_h + 1)$	–	6
P-PID	K_{p1}	$K_{p2} + K_{i2}/s + K_{d2}s/(s/\omega_h + 1)$	8
PI-(1+FOPID)	$K_{p1} + K_{i1}/s$	$1 + K_{p2} + K_{i2}s^{-\lambda} + K_{d2}s^{\mu}$	14
FOPID-FOPID	$K_{p1} + K_{i1}s^{-\lambda_1} + K_{d1}s^{\mu_1}$	$K_{p2} + K_{i2}s^{-\lambda_2}$	16
FOPID-FOPD	$K_{p1} + K_{i1}s^{-\lambda}$	$K_{p2} + K_{d2}s^{\mu}$	12

derivatives use the same high-frequency limit, $K_d s/(s/\omega_h + 1)$ with $\omega_h = 100$ rad/s. The reference passes a critically damped second-order model with $\omega = 1$ rad/s, identical for all structures.

4. Tuning problem

4.1. Cost

Two training cases are used, whose disturbance and noise realizations differ from those of the tests: a simultaneous step in x , y , z and ψ (20 s), and station keeping in waves and current with steps in ψ and z (30 s). The cost is

$$J = \sum_s \left[\sum_{i=1}^6 w_i \int_0^{T_s} t |e_i(t)| dt + \rho E_s \right] + K_{pm} \sum_a \sum_{i=1}^6 \frac{\max(0, \text{PM}_a^* - \text{PM}_{a,i})}{\text{PM}_a^*}, \quad (5)$$

with $w = [1, 1, 1, 2, 2, 2]$ (one metre of error weighs as much as 0.5 rad), the electrical energy E_s in joules, $\rho = 2 \times 10^{-4} \text{ J}^{-1}$ and $K_{pm} = 200$. The margins are required at three actuator points a : the nominal thruster ($\tau_m = 0.1$ s,

30 ms; $\text{PM}^* = 45^\circ$) and the slow (0.2 s, 60 ms) and fast (0.05 s, 10 ms) corners ($\text{PM}^* = 20^\circ$). Diverging runs receive $10^4(1 + \text{lost fraction})$.

4.2. Margin constraint

About hover, DOF i is $\ddot{y} + c_{1,i}\dot{y} + c_{0,i}y = H(s)a$ with $c_1 = D_{l,ii}/M_{ii}$ and the restoring stiffness $c_0 = z_b B/M_{ii}$ in roll and pitch. The actuator path $H(s) = e^{-(n_d+1/2)\Delta t s}/(\tau_m s + 1)$ contains n_d periods of latency, the zero-order hold and the thruster lag. Broken at a , the loop is

$$L_i(s) = G_2(s)(G_1(s) + s)P_i(s), \quad P_i(s) = \frac{H(s)}{s^2 + c_{1,i}s + c_{0,i}}, \quad (6)$$

($L_i = G_1 P_i$ for single-loop structures), with G_1, G_2 the stage transfer functions as realized (Oustaloup). PM_i is the smallest distance of $\arg L_i$ from -180° over all gain crossovers below 300 rad/s; a loop whose crossover lies above (the Nyquist frequency is 314 rad/s) gets $\text{PM}_i = -180^\circ$. The linear analysis predicts the latency at which the simulated loop loses stability (unit test `test_margins_predict_simulated_delay_limit`).

4.3. Search space

Gains are searched as $\log_{10} K \in [-2, \log_{10} 50]$, the orders as $\lambda \in [0, 1.5]$, $\mu \in [0, 1]$, and $n \in [1, 10]$. On a linear gain scale most random initial points violate the margin constraint and all tuners stalled near $J \approx 100$.

5. Tuners and comparison protocol

5.1. Stellar oscillation optimizer

SOO [6] moves a population of N oscillators in two stages per iteration $t = 1, \dots, T$. With the angular frequency $\omega(t) = 2\pi/(P_0 + \Delta P t)$, the scaling

factor $S(t) = 2(1 - t/T)$, the best position $\mathbf{x}_b$ and $r_1, r_2, r_3 \sim U(0, 1)$ drawn per component, the first stage forms

$$\mathbf{x}_{o1} = \mathbf{x}_b - r_1 r_3 S(\omega r_1 - 1) (\mathbf{x}_i - |r_1 \sin(r_2) |r_3 \mathbf{x}_b|), \quad (7)$$

$$\mathbf{x}_{o2} = \mathbf{x}_b - r_2 r_3 S(\omega r_1 - 1) (\mathbf{x}_i - |r_1 \cos(r_2) |r_3 \mathbf{x}_b|), \quad (8)$$

$$\mathbf{x}_{new} = r_3 (\mathbf{x}_{o1} + \mathbf{x}_{o2})/2, \quad (9)$$

and the second stage moves each oscillator about the mean $\bar{\mathbf{x}}_3$ of the three best,

$$\mathbf{x}_{osc} = \bar{\mathbf{x}}_3 + \frac{1}{2} [\sin(r\pi)(\mathbf{x}_a - \mathbf{x}_{b'}) + \cos((1-r)\pi)(\mathbf{x}_a - \mathbf{x}_c)], \quad (10)$$

with three other oscillators a, b', c , keeping each component of the old position with probability $1/2$. We follow the equations of the paper as read by the reimplementation in [9]: both stages in every iteration with greedy survivor selection, $P_0 = 3$, $\Delta P = 0.001$, clipping at the bounds. That source documents several differences between the paper and the authors' released code. SOO therefore spends $2N$ evaluations per iteration.

Eqs. (7)–(9) refer to the coordinate origin twice: through $|r_3 \mathbf{x}_b|$ and through the factor $r_3 \in [0, 1]$, which contracts every candidate towards $\mathbf{0}$. As a control we use *BC-SOO*, the same first stage written in coordinates relative to $\mathbf{x}_b$ (the substitution $\mathbf{x} \mapsto \mathbf{x} - \mathbf{x}_b$),

$$\begin{aligned} \mathbf{x}_{ok} &= \mathbf{x}_b - r_k r_3 S(\omega r_1 - 1) (\mathbf{x}_i - \mathbf{x}_b), \quad k = 1, 2, \\ \mathbf{x}_{new} &= \mathbf{x}_b + r_3 ((\mathbf{x}_{o1} + \mathbf{x}_{o2})/2 - \mathbf{x}_b), \end{aligned} \quad (11)$$

which is translation invariant by construction.

5.2. Other tuners

PSO with the constriction coefficients of [14] ($w = 0.7298$, $c_1 = c_2 = 1.49618$, velocity limited to 20% of the range), DE/rand/1/bin [4] ($F = 0.5$, $CR = 0.9$), GWO [15], GJO [16] and uniform random search (RS). PSO and DE are the tuners of [1]; GWO and GJO are among the comparison methods of [5].

5.3. Protocol

- *Equal budget.* Every run stops after 3030 evaluations (population 30): 100 iterations for the one-stage methods and 50 for SOO. The equal-iteration protocol of [5], which gives SOO 6030 evaluations, is run as a separate setting (SOO-iter).
- *Mirrored box.* Every tuner is also run on the problem expressed in the coordinates $\mathbf{x}' = \mathbf{lb} + \mathbf{ub} - \mathbf{x}$. The problem is the same; only the position of the coordinate origin relative to the good region changes. A tuner without positional bias gives the same distribution of results in both boxes.
- *Shifted benchmarks.* Sphere, Rastrigin, Rosenbrock and Ackley functions in $D = 22$ on $[-100, 100]^{22}$, with the optimum at the origin and at a random point of $[-60, 60]^{22}$ (15 runs).
- Ten independent runs per setting; medians, interquartile ranges and two-sided Mann–Whitney tests.

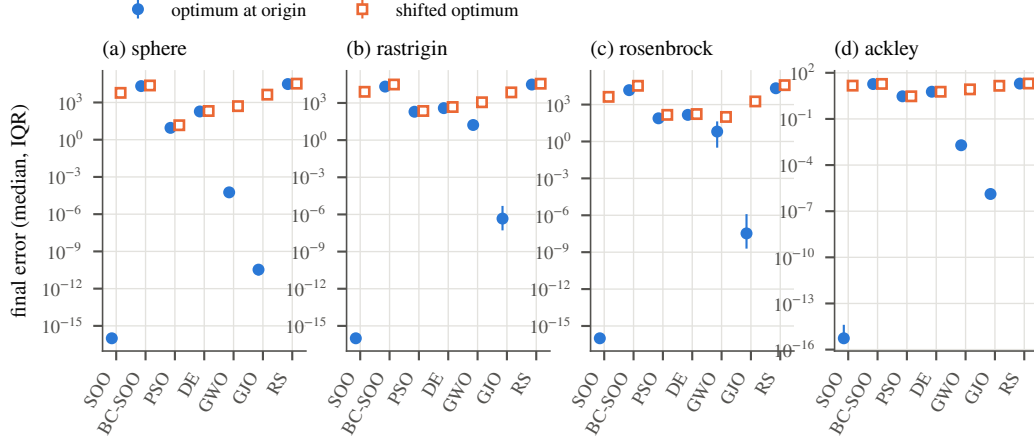


Figure 1: Final error after 3030 evaluations ($D = 22$, box $[-100, 100]^{22}$, 15 runs; median and interquartile range) with the optimum at the origin (filled) and at a random point in $[-60, 60]^{22}$ (open). Values below 10^{-15} are drawn at 10^{-16} .

6. Results

6.1. Positional bias on shifted benchmarks

With the optimum at the origin, SOO reaches 10^{-49} on the sphere and exactly zero on Rastrigin and Rosenbrock; with the optimum shifted it ends at 6×10^3 , 8×10^3 and 4×10^3 (Fig. 1). GJO and GWO show the same pattern with smaller gaps, while PSO and DE give the same errors in both cases, as expected from their translation-invariant updates. BC-SOO is translation invariant but performs like random search: the search ability of SOO comes from its attraction to the origin.

6.2. Tuners on the six-DOF problem

PSO gives the lowest median cost and the same distribution in both boxes (Table 3, Fig. 2). SOO is significantly worse than PSO at the same budget;

Table 3: Training cost J of the proposed controller for ten runs per tuner (3030 evaluations unless noted). p : two-sided Mann–Whitney tests against PSO (original box) and between the two boxes.

tuner	best	median [IQR], original box	median [IQR], mirrored box	p vs PSO	p boxes
PSO	35.8	40.0 [37.9, 56.2]	40.4 [38.9, 46.1]	–	1.000
DE	54.7	67.0 [58.3, 73.3]	84.4 [56.4, 99.8]	0.003	0.273
GWO	48.5	68.4 [55.9, 113.9]	103.8 [84.9, 119.1]	0.007	0.186
SOO	41.5	85.0 [69.6, 126.4]	59.7 [52.6, 105.3]	0.002	0.385
SOO, 6030 eval.	32.1	60.7 [43.4, 80.3]	–	0.089	–
GJO	57.5	122.3 [90.7, 142.0]	208.0 [165.8, 228.6]	< 0.001	0.002
BC-SOO	215.8	329.8 [275.1, 410.9]	325.3 [186.1, 435.1]	< 0.001	0.521
RS	193.7	403.8 [279.5, 496.1]	363.1 [321.8, 412.6]	< 0.001	0.623

with the equal-iteration protocol, which doubles its evaluations, the difference is no longer significant ($p = 0.089$) but its median stays 50% higher. GJO depends significantly on the box ($p = 0.002$): its results change when only the position of the coordinate origin relative to the good region changes. On this problem the origin of the search box is not near the optimum (the gains are searched as $\log_{10} K$), so the attraction that makes SOO excel on centred benchmarks no longer helps. PSO is used for the comparison of structures below; DE is kept as the second tuner because it was the best tuner in [1].

6.3. Controller structures

The single-loop structures (FOPID, PID) and FOPID-FOPI cannot meet the margin constraints with useful bandwidth: their training costs are an order of magnitude higher (Table 4) and the vehicle oscillates in waves (Fig. 4). Among the cascades with a derivative action in the inner stage, the integer P-PID reaches the lowest nominal training cost, and the proposed FOPID-(1+TFOID) the second lowest median. The picture changes in the test cases

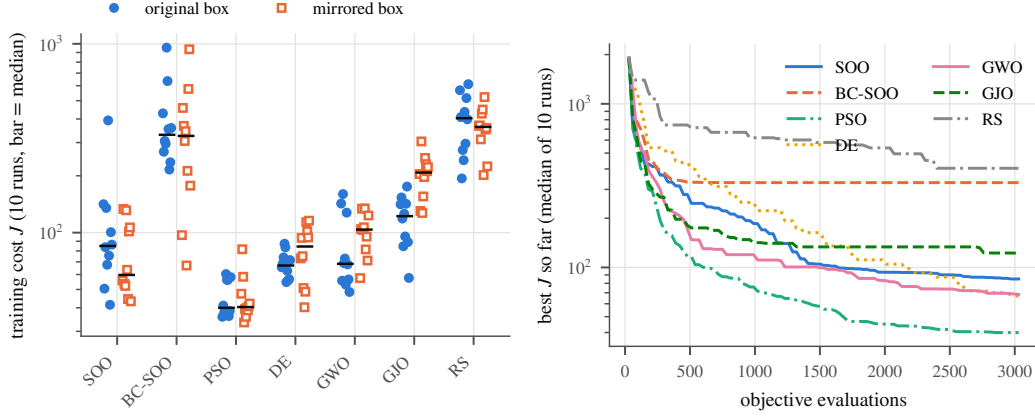


Figure 2: Left: training cost of every run in the original and mirrored boxes (bar: median). Right: median best-so-far cost against the number of evaluations (original box).

(Table 5, Fig. 3). The proposed controller has the lowest error in T4, 6.5 times less error than P-PID and 8.6 times less than PI-(1+FOPID) under severe white noise (T2), and no divergence in the Monte Carlo study, whose 90th percentile is also the lowest. FOPI-FOPD is better in T1 and T2 and within 0.5% in T4, but it diverges once in the Monte Carlo study and its 90th percentile is 24% higher. P-PID has the lowest error only in the storm test T3, 10% below the proposed controller. The energy in T4 is the same for all cascades within 3%.

6.4. Robustness

Mass, added mass, damping and buoyancy changes move the cost of the cascades by at most 11% (Fig. 5). A 12 V battery, which lowers the maximum thrust by 28–29% on the T200 curves, costs 22–40%; the slow thruster with doubled latency 24–57%. Losing a horizontal thruster without telling the allocation makes every cascade diverge, and losing a vertical one costs 4.4–12

Table 4: Training cost of the seven structures over ten runs per tuner.

structure	DE		PSO	
	best	median	best	median
P-PID	33.9	36.2	33.4	70.1
FOPID-(1+TFOID)	54.7	67.0	35.8	40.0
PI-(1+FOPID)	37.6	51.6	33.7	52.9
FOPI-FOPD	45.8	57.9	38.4	56.2
FOPID-FOPI	327.7	373.3	309.1	315.6
FOPID	377.6	402.0	372.2	388.9
PID	479.8	639.6	479.9	639.6

times; FOPID, FOPID-FOPI and PID, which are less aggressive, degrade by 2–10 times (PID diverges in T4). When the failure is known and the allocation is recomputed on the remaining seven thrusters, the cost of the proposed controller rises by only 10–13% (P-PID 10–23%): the over-actuation of the heavy configuration is useful only together with fault-aware allocation.

6.5. Margins and latency

At the nominal thruster the six loops of the proposed controller have phase margins of 45.1–52.1° and tolerate 40–67 ms of additional latency; at the slow corner the smallest margin is 20.0°, on the constraint. To show what the constraint buys, the proposed controller was tuned as in the load-frequency study: no latency in the model and no margin requirement (X5, six runs). The training cost drops to 18.6–22.2, but the smallest margin per run is 3.8–24.7°. In T4 these controllers reach 75–99 without latency, 180–466 at the nominal 30 ms and 422–1314 at 60 ms, while the margin-constrained

Table 5: Test cases for the best tuned controller of each structure (lowest training cost). Weighted ITAE as in Eq. (5); E : electrical energy in T4 (60 s); Monte Carlo: T4 with random mass ($\pm 20\%$), added mass and damping ($\pm 50\%$), buoyancy (-2 to $+3\%$), battery 12–20 V, thruster lag 0.05–0.2 s and latency 10–60 ms.

structure (tuner)	weighted ITAE				E T4	Monte Carlo, T4 (100)		
	T1	T2	T3	T4	[kJ]	median	p90	diverged
FOPID-(1+TFOID) (PSO)	1.7	66.4	52.4	128.8	11.4	146	275	0
FOPI-FOPD (PSO)	1.4	29.8	52.3	129.4	11.7	148	340	1
PI-(1+FOPID) (PSO)	2.6	572.1	48.9	131.9	11.7	146	547	12
P-PID (PSO)	3.1	432.3	47.3	152.8	11.4	166	493	8
FOPID-FOPI (PSO)	31.9	45.7	400.6	1097.1	9.5	1304	2545	1
FOPID (PSO)	20.5	415.3	475.9	1274.4	9.8	1597	3462	3
PID (DE)	31.7	688.9	519.1	1385.0	9.7	1927	7518	3

controller goes from 128.6 to 135.3 over the same range. The lower nominal cost of unconstrained tuning is an artefact of the missing latency.

7. Discussion and limitations

- *What the fractional cascade buys.* Under realistic latency and margin requirements its benefit is robustness (noise, parameter spread, no divergence) rather than nominal performance; the integer P-PID cascade is equally good or better on the nominal training cases.
- *Tuning budget.* The proposed structure has 22 parameters against 8 for P-PID. It contains an approximation of P-PID, so its optimum should not be worse; the higher best cost after 3030 evaluations points to a search limitation.

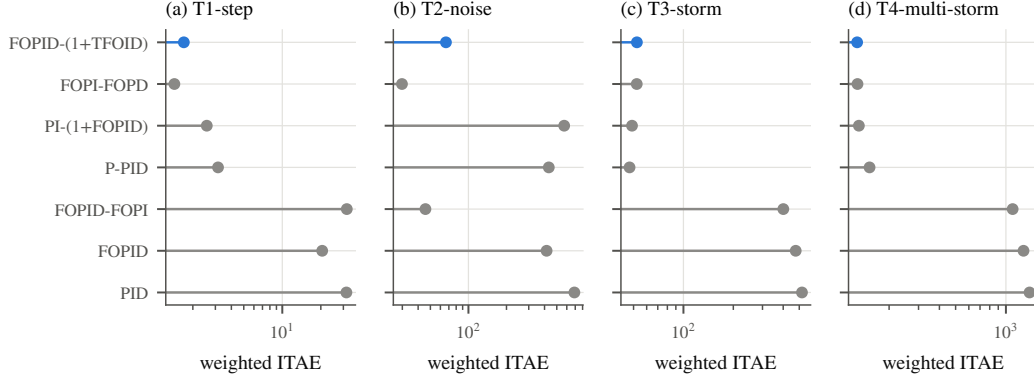


Figure 3: Weighted ITAE in the four test cases (logarithmic axes).

- *Thruster dynamics.* No validated dynamic model of the T200 was available; the design covers a range of lag and latency instead. [TODO: Replace the nominal values by those of the MATLAB re-identification and narrow the range accordingly.]
- *Model.* Simulation only; hydrodynamic coefficients from [12]; one parameter set per group of DOFs; margins from a linearization about hover; navigation errors modelled, not estimated. Pool tests or hardware-in-the-loop, as in the FPGA realization of [1], are the next step.

8. Conclusions

[TODO: Three to four sentences: the cascade in six DOF with fitted T200 curves; margin-constrained tuning over an actuator range; tuners compared at equal budget and in a mirrored box (PSO best, SOO and GJO biased); the proposed cascade is the most robust structure but not better than P-PID nominally; fault-aware allocation.]

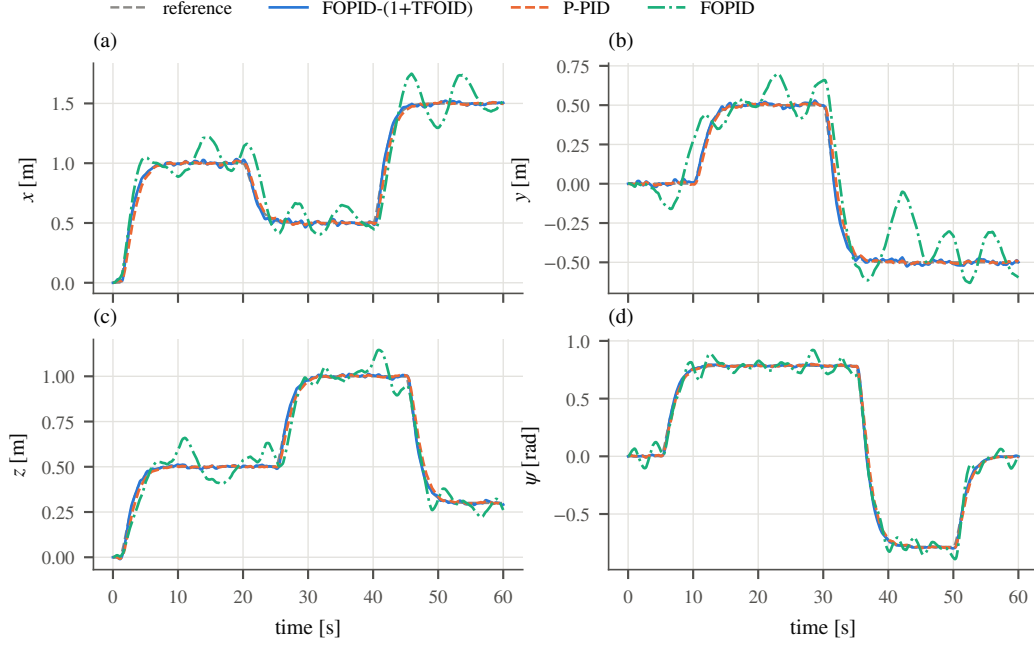


Figure 4: Test T4 (multi-level set points in waves and current): x , y , z and ψ for the proposed cascade, the integer P-PID cascade and the single-loop FOPID of [1], each with its best tuning.

Data and code availability

Code, results and figure scripts: [\[TODO: repository URL\]](#). The T200 data are fetched from the manufacturer and checked by SHA-256 [10].

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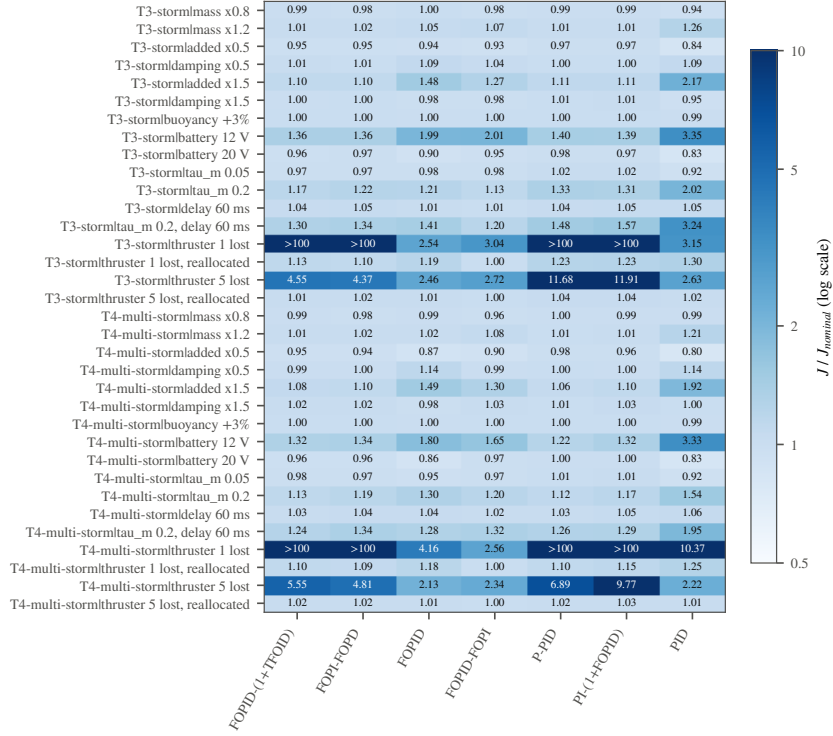


Figure 5: Cost of each variant relative to the nominal test case, per structure (the controllers are not retuned). Ratios are relative to each structure’s own nominal cost; the single-loop structures start from a much higher nominal cost (Table 5).

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